MDM: Financial Technology (S.Y. SEM: III)

Chapter No: 1 Introduction to Financial Systems

Notes

Q1. Define a Financial System. Explain its objectives and importance in the economic development of a country.

A Financial System is a network of financial institutions, financial markets, financial instruments, and financial services that helps in the movement of money from people who have surplus funds to people who need funds. It supports saving, borrowing, lending, investing, and making payments in the economy.

Objectives of a Financial System

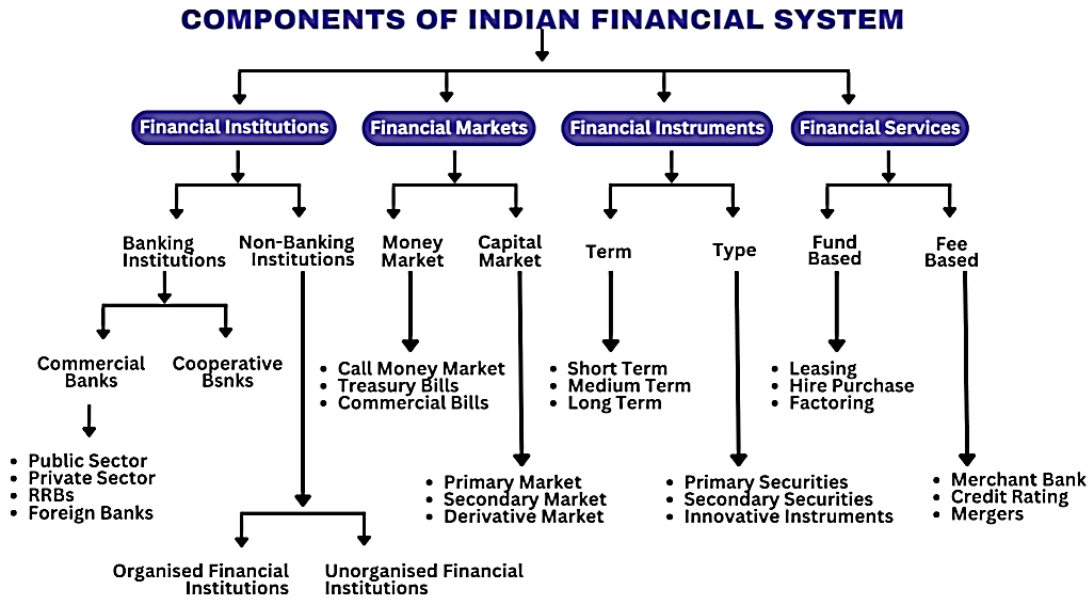
1. *Mobilization of Savings:* The financial system collects savings from individuals and organizations and channels these funds towards productive investments in different sectors of the economy.
2. *Allocation of Funds:* The financial system transfers money from savers to businesses, industries, and governments that require funds for various economic activities.
3. *Capital Formation:* The financial system helps businesses raise capital for expansion, modernization, and the establishment of new industries.
4. *Providing Liquidity:* The financial system allows investors to convert their financial assets into cash easily whenever they require money.
5. *Risk Management:* The financial system provides insurance services and different investment options that help individuals and businesses reduce financial risk.
6. *Efficient Payment System:* The financial system enables safe and fast financial transactions through services such as UPI, NEFT, RTGS, debit cards, and credit cards.

Importance in Economic Development

1. The financial system encourages investment by making funds available to businesses and industries.
2. It supports industrial and infrastructure development by providing long-term and short-term finance.
3. It creates employment opportunities because new businesses and industries require workers.
4. It promotes entrepreneurship by providing loans and financial assistance to new business owners.
5. It increases the production of goods and services, which contributes to the country's economic growth.
6. It improves the standard of living by increasing income, employment, and access to financial services.

Q2. Draw and explain the structure of the Financial System, highlighting Financial Markets, Financial Institutions, Financial Instruments, and Financial Services.

The Financial System consists of four major components that work together to ensure the smooth flow of money in the economy.



1. Financial Institutions

Financial Institutions are organizations that collect savings and provide loans and other financial services.

Examples include the Reserve Bank of India (RBI), State Bank of India (SBI), HDFC Bank, LIC, and NBFCs.

These institutions promote savings, provide credit, and support economic development.

2. Financial Markets

Financial Markets are places where buyers and sellers trade financial assets such as shares, bonds, and money market instruments.

Examples include the Capital Market, Money Market, and Derivatives Market.

These markets help companies and governments raise funds while providing investment opportunities to individuals.

3. Financial Instruments

Financial Instruments are financial assets or documents that represent ownership or debt.

Examples include shares, debentures, bonds, treasury bills, certificates of deposit, and mutual fund units.

These instruments help businesses raise funds and provide investment opportunities to investors.

4. Financial Services

Financial Services are services provided by financial institutions to help people and businesses manage their money.

Examples include banking, insurance, investment management, digital payments, stock broking, and financial advisory services.

These services make financial transactions safe, convenient, and efficient.

Q3. Explain the different types of Financial Markets. Discuss the Capital Market, Money Market, and Derivatives Market with suitable examples.

Financial Markets are places where financial assets are bought and sold. They help in transferring funds from people who have surplus money to people and organizations that require funds.

1. Capital Market

The Capital Market deals with long-term funds where the maturity period is more than one year.
It helps companies and governments raise money for expansion, infrastructure projects, and business development.
The main instruments traded in the capital market are equity shares, preference shares, debentures, and bonds.
The major participants include companies, investors, mutual funds, banks, and stock exchanges.

Example: A company raises money by issuing shares through an Initial Public Offering (IPO).

2. Money Market

The Money Market deals with short-term funds where the maturity period is less than one year.
It helps banks, financial institutions, companies, and governments meet their short-term financial requirements.
The main instruments traded in the money market are Treasury Bills, Commercial Papers, Certificates of Deposit, and Call Money.
The Reserve Bank of India, commercial banks, and financial institutions are the major participants in this market.

Example: A commercial bank invests in a 91-day Treasury Bill issued by the Government of India.

3. Derivatives Market

The Derivatives Market deals with financial contracts whose value depends on another asset such as shares, commodities, currencies, or interest rates.
It is mainly used to reduce financial risk and protect investors against changes in market prices.
The common derivative instruments include futures, options, forwards, and swaps.
The major participants include traders, investors, banks, companies, and financial institutions.

Example: An exporter uses a currency futures contract to protect against fluctuations in foreign exchange rates.

Q4. Compare the Capital Market, Money Market, and Derivatives Market based on objectives, instruments, maturity, participants, and risk.

Basis	Capital Market	Money Market	Derivatives Market
Objective	The Capital Market helps companies and governments raise long-term funds for business growth and development.	The Money Market helps institutions meet their short-term financial requirements and maintain liquidity.	The Derivatives Market helps investors manage financial risk and protect against price fluctuations.
Instruments	The Capital Market deals with shares, debentures, and bonds.	The Money Market deals with Treasury Bills, Commercial Papers, Certificates of Deposit, and Call Money.	The Derivatives Market deals with futures, options, forwards, and swaps.
Maturity	The maturity period is more than one year.	The maturity period is less than one year.	The maturity depends on the expiry date mentioned in the contract.
Participants	Companies, investors, banks, mutual funds, and stock exchanges participate in the Capital Market.	The Reserve Bank of India, commercial banks, financial institutions, and the government participate in the Money Market.	Traders, investors, banks, companies, and financial institutions participate in the Derivatives Market.
Risk	The Capital Market has moderate to high risk because security prices change frequently.	The Money Market has low risk because it deals with short-term financial instruments.	The Derivatives Market has high risk because the value of contracts depends on changes in the underlying asset.

Q5. What are Financial Institutions? Explain their types and functions with suitable examples.

Financial Institutions are organizations that collect money from savers and provide financial services such as loans, investments, insurance, and payment facilities. They help in the efficient movement of funds within the economy.

Types of Financial Institutions

1. Banking Institutions

Banking institutions accept deposits from customers and provide loans to individuals, businesses, and governments. They also offer services such as money transfer, internet banking, and foreign exchange facilities.

Examples include the Reserve Bank of India (RBI), State Bank of India (SBI), HDFC Bank, and Bank of Baroda.

2. Non-Banking Financial Institutions (NBFIs)

Non-Banking Financial Institutions provide financial services such as loans, insurance, leasing, and investment services but generally do not accept demand deposits like commercial banks.

Examples include LIC, Bajaj Finance, Tata Capital, and mutual fund companies.

Functions of Financial Institutions

1. *Accepting Deposits:* Financial institutions encourage people to save money by offering different types of deposit schemes.
2. *Providing Loans:* Financial institutions provide loans for education, housing, agriculture, business, and personal needs.
3. *Promoting Investments:* Financial institutions offer investment products such as mutual funds, bonds, fixed deposits, and government securities.
4. *Providing Payment Services:* Financial institutions enable safe and quick financial transactions through UPI, NEFT, RTGS, IMPS, debit cards, and credit cards.
5. *Supporting Financial Inclusion:* Financial institutions extend banking and financial services to rural and remote areas so that more people can participate in the formal financial system.
6. *Supporting Economic Development:* Financial institutions provide financial assistance to industries, businesses, infrastructure projects, and entrepreneurs, which helps generate employment and promotes economic growth.

Q.6 Differentiate Between Banking and Non-Banking Financial Institutions

Banking Financial Institutions (BFIs)	Non-Banking Financial Institutions (NBFIs)
1. Banking financial institutions accept public deposits such as savings accounts, current accounts, and fixed deposits from customers.	1. Non-banking financial institutions generally do not accept demand deposits such as savings and current accounts from the public.
2. They provide complete banking services, including accepting deposits, providing loans, fund transfers, and payment facilities.	2. They provide specialized financial services such as loans, insurance, mutual funds, leasing, housing finance, and investment services.
3. They create credit by lending money collected from customer deposits.	3. They provide credit mainly through their own funds or borrowed funds and generally do not create credit like banks.
4. They operate the country's payment and settlement system through services such as NEFT, RTGS, IMPS, UPI, cheque clearing, and ATM networks.	4. They do not operate the national payment and settlement system like commercial banks.
5. They are mainly regulated by the Reserve Bank of India (RBI).	5. They are regulated by different authorities depending on their activities, such as the RBI, SEBI, IRDAI, and PFRDA.

6. Their primary objective is to provide comprehensive banking services, mobilize savings, and support economic growth.	6. Their primary objective is to provide specialized financial services and improve access to finance in different sectors of the economy.
7. Examples include State Bank of India (SBI), HDFC Bank, ICICI Bank, Punjab National Bank (PNB), and Bank of Baroda.	7. Examples include Bajaj Finance, LIC, SBI Mutual Fund, LIC Housing Finance, Tata Capital, and the National Pension System (NPS).

Q7. Explain the different types of Financial Instruments. Discuss Equity Shares, Bonds, and Derivatives with suitable examples.

Financial Instruments are financial assets or legal documents that represent ownership, debt, or a financial contract. They help individuals, businesses, and governments raise funds and provide investment opportunities to investors.

Types of Financial Instruments

1. Equity Shares

Equity Shares represent the ownership of a company. A person who purchases equity shares becomes a part owner or shareholder of the company.

Shareholders receive dividends if the company earns profits and may also benefit from an increase in the share price. However, dividends are not guaranteed because they depend on the company's performance.

Example: If an investor purchases shares of Reliance Industries Ltd., the investor becomes a shareholder and can earn dividends and capital appreciation.

2. Bonds

Bonds are debt instruments issued by governments or companies to raise money. When an investor purchases a bond, the investor lends money to the issuer for a fixed period.

In return, the issuer pays a fixed rate of interest called a coupon and repays the principal amount on maturity.

Example: Government of India Bonds, Corporate Bonds issued by companies such as NTPC or REC.

3. Derivatives

Derivatives are financial contracts whose value depends on the value of another asset called the underlying asset.

The underlying asset may be shares, commodities, currencies, or interest rates. Derivatives are mainly used for risk management and hedging against price fluctuations.

Common derivatives include Futures, Options, Forwards, and Swaps.

Example: An exporter purchases a currency futures contract to protect against changes in foreign exchange rates.

Importance of Financial Instruments

1. Financial instruments help businesses and governments raise funds for various projects.
2. They provide different investment opportunities to individuals and institutions.
3. They help investors earn income through dividends, interest, or capital appreciation.
4. They improve liquidity by allowing investors to buy and sell financial assets.
5. Derivatives help businesses and investors reduce financial risk caused by changes in market prices.

Q8. Differentiate between Equity Shares, Bonds, and Derivatives

Basis	Equity Shares	Bonds	Derivatives
Ownership	Equity shares represent ownership in a company, and shareholders become part owners of the business.	Bonds do not provide ownership because they represent a loan given to the issuer.	Derivatives do not provide ownership because they are financial contracts based on another asset.
Return	Investors earn returns through dividends and an increase in the market value of shares.	Investors receive fixed interest payments and repayment of the principal amount on maturity.	Returns depend on changes in the price of the underlying asset and market conditions.
Risk	Equity shares carry high risk because share prices change frequently.	Bonds have relatively lower risk because interest payments are generally fixed.	Derivatives involve high risk because their value changes rapidly with market prices.
Maturity	Equity shares generally do not have a maturity period.	Bonds have a fixed maturity period after which the principal amount is repaid.	Derivatives have a fixed expiry date mentioned in the contract.
Purpose	Equity shares help companies raise long-term capital for business expansion.	Bonds help governments and companies borrow money for a specific period.	Derivatives are mainly used for hedging risk and managing price fluctuations.

Q9. Explain the role of the Financial System in the economic development of India.

The Financial System plays an important role in the economic development of India by ensuring the efficient flow of money between savers and borrowers. It supports investment, production, employment, and overall economic growth.

Role of the Financial System

1. *Mobilization of Savings:* The financial system encourages individuals and organizations to save money through banks, post offices, and other financial institutions. These savings are used for productive investments.
2. *Capital Formation:* The financial system helps businesses and industries raise funds for expansion and modernization, which increases capital formation in the country.
3. *Industrial Development:* Financial institutions provide loans and financial assistance to industries, enabling them to establish new projects and increase production.
4. *Employment Generation:* When industries receive financial support, they expand their operations and create new employment opportunities.
5. *Promotion of Investment:* The financial system offers various investment options such as shares, bonds, mutual funds, and fixed deposits, encouraging people to invest their money.
6. *Financial Inclusion:* Banks and digital financial services provide access to banking facilities in rural and remote areas, helping more people participate in the formal economy.
7. *Support to Government Projects:* The financial system helps the government raise funds for infrastructure projects, education, healthcare, transportation, and other public welfare activities.
8. *Economic Stability:* The Reserve Bank of India regulates the financial system and maintains monetary stability by controlling inflation and ensuring sufficient liquidity in the economy.

Q10. Explain the functions, objectives, and importance of the Reserve Bank of India (RBI) in regulating the Indian financial system.

The Reserve Bank of India (RBI) is the central bank of India. It was established on 1 April 1935 under the Reserve Bank of India Act, 1934. The RBI regulates and supervises the Indian financial system and ensures monetary and financial stability.

Objectives of RBI

1. The RBI aims to maintain price stability by controlling inflation in the economy.
2. It ensures adequate availability of credit for economic growth and development.
3. It maintains financial stability by regulating banks and financial institutions.
4. It promotes a safe, efficient, and reliable payment system throughout the country.

Functions of RBI

1. *Issue of Currency***:** The RBI has the authority to issue and manage currency notes in India, except one-rupee notes and coins issued by the Government of India.
2. *Banker to the Government:* The RBI manages the banking transactions, public debt, and financial accounts of the Central and State Governments.
3. *Banker's Bank:* The RBI supervises commercial banks and provides financial assistance whenever required.
4. *Monetary Policy:* The RBI controls the supply of money and interest rates through monetary policy to maintain economic stability.
5. *Foreign Exchange Management:* The RBI manages India's foreign exchange reserves and regulates foreign exchange transactions under FEMA.
6. *Regulation of Banks:* The RBI grants banking licences, monitors bank performance, and ensures that banks follow financial regulations.

Importance of RBI

1. The RBI maintains stability in the banking system.
2. It controls inflation and supports sustainable economic growth.
3. It ensures the safety of depositors' money by regulating banks.
4. It promotes digital payment systems such as UPI, NEFT, RTGS, and IMPS.
5. It maintains confidence in the Indian financial system by ensuring transparency and financial discipline.

Q11. Explain the functions, powers, and objectives of the Securities and Exchange Board of India (SEBI). How does SEBI regulate the securities market and protect investors?

The Securities and Exchange Board of India (SEBI) is the regulator of the securities market in India. It was established in 1988 and was given statutory powers under the SEBI Act, 1992. SEBI regulates stock exchanges, protects investors, and promotes the development of the securities market.

Objectives of SEBI

1. SEBI protects the interests of investors in the securities market.
2. It promotes the orderly development of the capital market.
3. It ensures fair, transparent, and efficient trading practices.
4. It prevents unfair trade practices such as fraud and insider trading.

Functions of SEBI

1. *Regulating Stock Exchanges:* SEBI supervises stock exchanges such as the Bombay Stock Exchange (BSE) and the National Stock Exchange (NSE).

2. *Registration of Market Participants:* SEBI registers and regulates brokers, merchant bankers, mutual funds, portfolio managers, and other market intermediaries.
3. *Monitoring Public Issues:* SEBI regulates Initial Public Offerings (IPOs) and ensures that companies provide accurate information to investors.
4. *Investor Education:* SEBI conducts awareness programs to educate investors about their rights and investment risks.
5. *Inspection and Investigation:* SEBI investigates market irregularities and takes action against individuals or organizations that violate securities laws.

Powers of SEBI

1. SEBI has the power to inspect the records of stock exchanges and market intermediaries.
2. It can impose penalties and fines on organizations or individuals who violate securities regulations.
3. It can suspend or cancel the registration of brokers and other market participants.
4. It can prohibit insider trading, price manipulation, and other fraudulent activities.

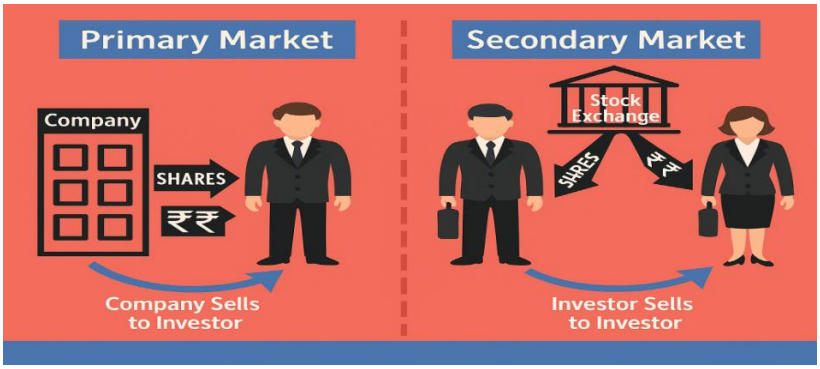
How SEBI Regulates the Securities Market and Protects Investors

1. SEBI makes rules and regulations to ensure fair and transparent trading in the securities market.
2. It monitors the activities of stock exchanges and market intermediaries to prevent market manipulation.
3. It takes strict action against fraudulent practices such as insider trading and misleading advertisements.
4. It provides a grievance redressal mechanism to help investors resolve complaints against listed companies and market intermediaries.
5. It promotes investor awareness through educational programs, guidelines, and online resources to encourage informed investment decisions.

Explain how capital formation takes place through primary & secondary equity market ?

Capital formation is the process of raising capital in an economy. In the equity market, this happens when savings from investors turn into long-term funds for companies.

- **Capital Formation in the Primary Market**: The primary market is where companies sell brand-new shares of stock to the public for the first time.
  - Direct Flow of Funds: Money goes straight from the investor to the company.
  - Initial Public Offering (IPO): A private company lists its shares publicly to raise fresh money.
  - Follow-on Public Offer (FPO) or Rights Issue: Existing public companies issue more new shares to raise extra money for new factories, equipment, or paying off debt.
  - Direct Impact: This creates brand-new productive capital. The company uses the cash to buy physical assets and expand its business.
- **Capital Formation in the Secondary Market**: The secondary market is the stock exchange where people buy and sell already-owned shares.
  - No Direct Cash to Company: When you buy a share from another investor on the stock market, the company gets no new cash.
  - Providing Liquidity: It lets investors turn their shares back into cash whenever they want.
  - Encouraging Future Primary Investments: People feel safer buying new shares in the primary market because they know they can easily sell them later in the secondary market.
  - Fair Pricing: It sets a clear market value for the company. If a company's stock price is high in the secondary market, it can easily return to the primary market later to issue more shares at a high price.



Question : Explain what mutual funds are , types of Mutal Funds and how they work ?

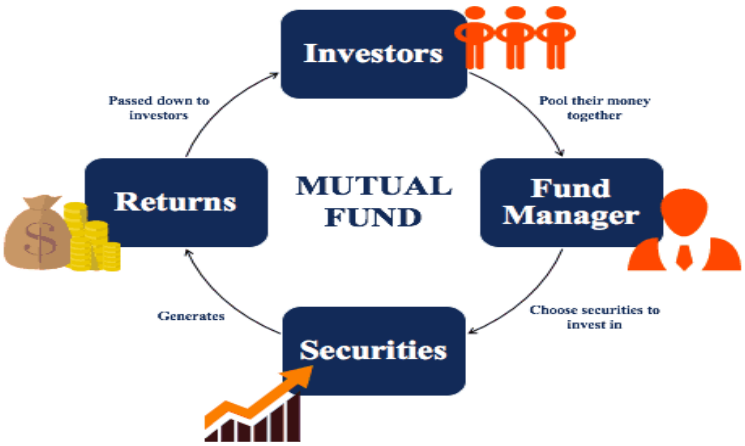
A mutual fund pools money from many people. Experts (Fund Managers) use this money to buy a mix of stocks, bonds, or other items. You own a small piece of the whole group instead of buying single company shares. This lowers your risk and saves you time.

Types of Mutual Funds:
Mutual funds are mainly split by asset class into equity funds (stocks), debt funds (bonds), hybrid funds (mix), and money market funds (short-term cash).
Equity Funds: Invest in company shares; aim for high long-term growth; carry higher risk.
Debt Funds: Invest in bonds and government securities; provide steady income; lower risk.
Hybrid Funds: Mix stocks and bonds; balance growth and stability; moderate risk.
Money Market Funds: Invest in ultra-short, safe cash tools; prioritize safety and high liquidity.

How Mutual Funds Works ?

- Pooling Money: Many people give cash to the fund.
- Buying a Basket: A fund manager uses the cash to buy many different assets.
- Getting Units: You get units or shares based on how much money you put in.

- Sharing Profit and Loss: If the assets grow in value, your shares grow too. If they drop, your shares drop.
- Daily Price: The value of one unit is called the Net Asset Value (NAV). It updates at the end of each business day.



Question: Explain what is Derivatives (Futures & Options) ?

Futures & Options
Futures and options (F&O) are financial "derivatives" that allow you to speculate on or hedge against the price of an asset (like stocks, commodities, or indices) without actually owning it.

Futures Contract :
A futures contract is a binding agreement between two parties to buy or sell an underlying asset at a predetermined price on a specific future date.
The Obligation: Both the buyer and the seller are bound by the contract to complete the transaction when the contract expires.
You buy a futures contract for a stock at ₹1,000. The stock goes up to ₹1,050 by the expiration date, you make a profit. However, if it falls to ₹950, you take a loss

Options Contract :
An options contract gives the buyer the right, but not the obligation, to buy or sell an underlying asset at a set price before or on a specific expiration date.
To get this flexibility, the buyer pays the seller a fee called a "premium"
Call Option: Gives you the right to buy. You buy this when you expect a stock's price to go up.
Put Option: Gives you the right to sell. You buy this when you expect a stock's price to go down



Mr. A buys a **call option** to buy a lot of 100 shares of Reliance Industries at the strike price (Agreed Price) of Rs. 1500 at a premium of Rs.2 per share. The contract expiry date is 30 days from now. Calculate Profit or loss if the actual price of Reliance Industries share is :
A: Rs.1650
B: Rs. 1300

Particulars	Scenario A	Scenario B
Contract Price	1500	1500
Market Price at Expiry	1650	1300
Will the option be exercised ?	Yes	No (No point in buying the share at Rs. 1500 when the Share is available in Market at Rs. 1300
No. of Shares	100	100
Premium paid	200	200
Profit = Change in Share Price Less Premium paid	=(1650-1500) X 100 Less 200 = **Rs. 14,800**	= 0- 200 =**- Rs. 200**



Question :What is Money Market and what are the types of Money Market Instruments ?

The money market in India is a wholesale financial market for short-term borrowing and lending, dealing exclusively in highly liquid debt instruments with maturities ranging from 1 day to 1 year. It is regulated primarily by the RBI. It provides a platform where banks, financial institutions, corporates, and the government can manage their immediate cash needs, deploy surplus funds, and maintain overnight liquidity

Types of Money Market Instruments

Type of Security	Issued By	Purpose	Maturity
Treasury bills (T-bills)	Central Government	To meet short-term fiscal deficits.	91 days, 182 days, & 364 days
Commercial paper (CP)	Corporates & Financial Institutions	To raise working capital	Between 7 days and 1 year.
Certificates of deposit (CD)	Scheduled commercial banks & financial institutions	To manage liquidity	7 days to 1 year
Call and notice money	Inter-bank	Short-term borrowing mechanisms.	Call money : 1 day Notice money : 2 & 14 days.

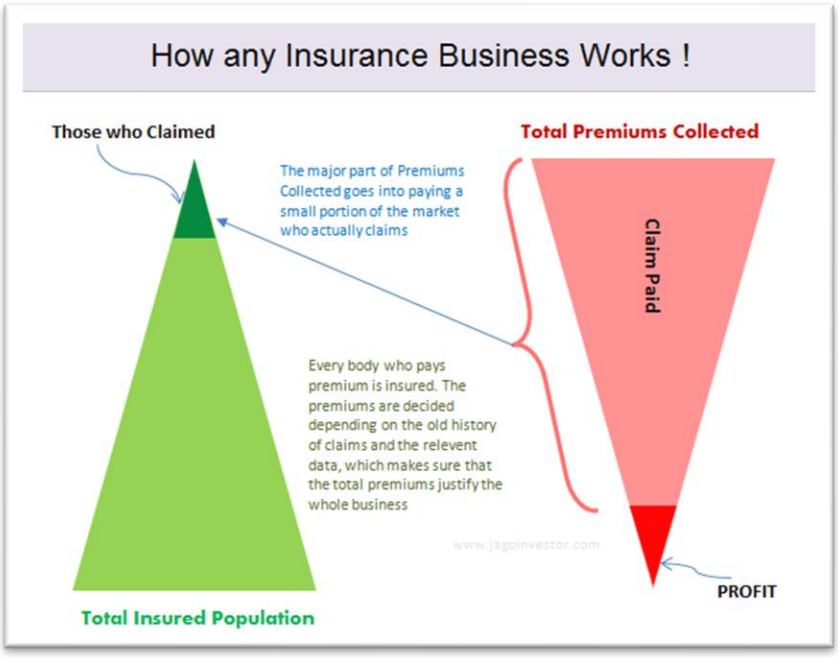
Question :What is the role and benefits of Insurance and what are the types of Insurance available ?

Insurance companies function to provide financial protection, pool collective risks, and manage investments. They achieve this by collecting regular payments, assessing potential dangers, and paying out valid claims when unexpected losses happen. These companies are regulated by the IRDAI (Insurance Regulatory and Development Authority of India)

Benefits of Insurance:

- **Risk Mitigation and Continuity** for Corporates: Business insurance safeguards companies against liability claims, operational disruptions, data breaches, and property losses, ensuring business continuity.
- **Financial Security for Families:** Life insurance delivers a payout (sum assured) to dependents to cover daily living costs, mortgage payments, and children's education if the policyholder passes away.
- **Medical Expense Management:** Health insurance covers hospitalisation costs, treatments, and critical illness management, keeping medical emergencies from draining personal savings.
- **Asset and Property Protection:** General insurance policies safeguard vehicles, homes, and personal belongings from theft, fire, natural disasters, or accidental damage.

Types of insurance:
Life insurance, health insurance, Travel insurance and property insurance (Motor , Home etc.) .



Question :Explain with examples what are fund based and fee based financial services ?

Fund-based financial services
Fund-based financial services involve the actual movement and lending of cash or physical capital, where institutions earn revenue through interest.

	A mutual fund pools money from many people to buy a mix of stocks, bonds, or other items. You own a small piece of the whole group instead of buying single company shares. This lowers your risk and saves you time
	Hire purchase is a transaction system where an asset is sold to a buyer who makes some part of payment as a down payment & agrees to pay the cost in regular monthly instalments.
	Leasing is a contractual arrangement where the owner of an asset (lessor) grants another party (lessee) the right to use the asset for a specific period in exchange for regular periodic payments (lease rentals).
	Venture Capital provides direct cash funding to startups in exchange for equity

Fee-Based Financial Services :
Fee-based (or non-fund based) services involve advisory, facilitation, or guarantee functions where no direct cash is lent, and institutions earn revenue through commissions, or service fees.

	Merchant banking is a specialized institutional financial service that acts as a corporate consultant for large enterprises for managing Initial Public Offerings (IPOs), Mergers and acquisitions (M&A) & Project counselling
	Stockbroking is a retail and institutional intermediary service that facilitates the buying and selling of securities on public stock exchanges like the NSE or BSE
	Underwriting is a financial risk-taking service where a financial institution (often a merchant bank or investment house) guarantees to buy any unsold shares or bonds during a public issue. They provide a financial safety net to companies raising capital.
	Credit rating is an independent assessment service that evaluates the creditworthiness and financial stability of a debt issuer (corporations).